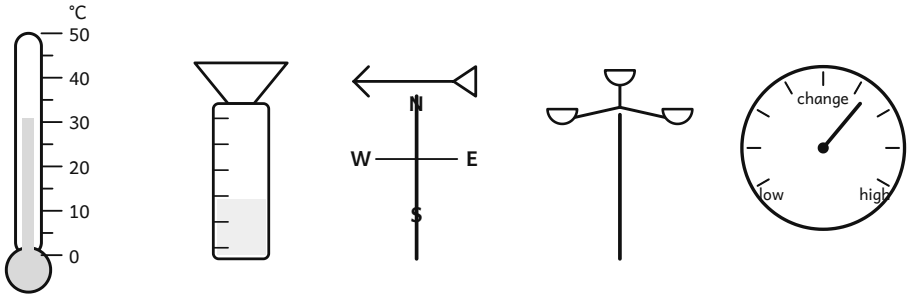
EARTH AND SPACE

Weather and Climate

Weather and climate, warm and cool places, moving air, weather instruments, clouds and forecasts



thermometer | rain gauge | wind vane | anemometer | barometer

In this pack

- **Learn it:** Weather, climate and how we measure them ★☆☆
- **Fact check:** Weather words and tools ★☆☆
- **Think it through:** Thinking about air and weather ★★☆
- **Investigate:** A week at our weather station ★★☆
- **Real-world case study:** Harvest before the flood: forecasts for haor farmers ★★★
- **Review and challenge:** Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Weather, climate and how we measure them

Name: Class: Date:

Learning goals

- Explain the difference between weather and climate.
- Explain how air moves when one place is warm and another is cool.
- Name weather instruments and cloud types, and use them to forecast the weather.

Will it rain on Sports Day? Should the farmer harvest the rice this week? Is a storm coming? People have always watched the sky to answer questions like these. Today, scientists called **meteorologists** use instruments, satellites and computers, but you can still be a weather detective with simple tools.

Weather or climate?

Weather is what the air is like **today or this week**: hot or cold, wet or dry, windy or calm, sunny or cloudy. It can change in a few hours. **Climate** is the **usual** weather of a place, worked out by measuring the weather for **many years** (usually 30).

> **Remember**
> "It is raining in Dhaka today" is **weather**. "Dhaka has hot, wet summers and cool, dry winters" is **climate**.

The climate of Bangladesh

Season	Months (about)	Usual weather
summer (pre-monsoon)	March – May	very hot; sudden storms (*kalbaishakhi*)
monsoon	June – October	warm, very wet, humid; most of the year's rain
winter	November – February	cool, dry, foggy mornings

Weather and climate affect **living things**: rice is planted when the monsoon rains come, many birds arrive in winter, and people wear lighter clothes in summer. They also affect **non-living things**: heavy rain makes rivers flood, and heat and rain slowly break up rocks and roads.

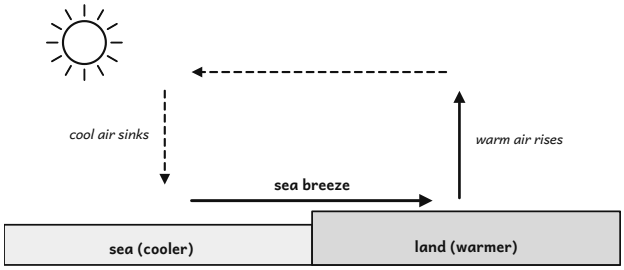
Warm places and cool places

Outdoors, it is warmest in **direct sunshine**, especially on dark surfaces like roads and tin roofs. It is coolest in the **shade** of trees and near water. Indoors, rooms with sunny windows and rooms under a tin roof are warmer. Warm air **rises**, so the air near the ceiling is warmer than near the floor.

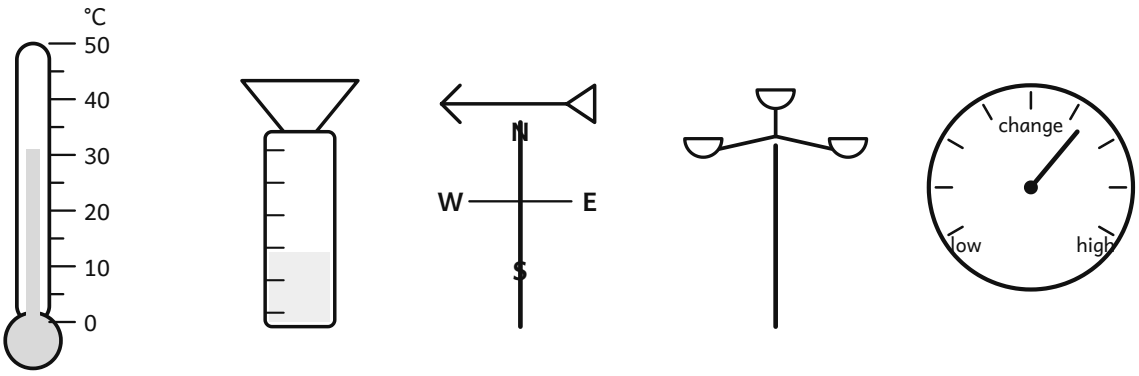
How air moves

When air is warmed it spreads out, becomes lighter and **rises**. Cooler air moves in to take its place. This moving air is **wind**. On a sunny day near the sea, the land warms up faster than the water, so a cool **sea breeze** blows from the sea to the land. At home, opening a low window and a high window lets cool air in and hot air out.

Measuring the weather



A sea breeze on a sunny day.



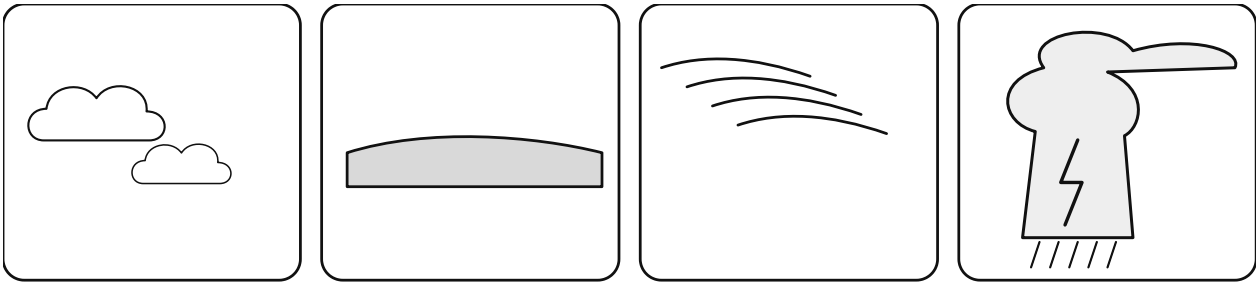
thermometer **rain gauge** **wind vane** **anemometer** **barometer**

Weather instruments.

Instrument	Measures	Unit or reading
thermometer	temperature	degrees Celsius (°C)
rain gauge	rainfall	millimetres (mm)
wind vane	wind direction (where the wind comes **from**)	N, S, E, W
anemometer	wind speed	km/h (or how fast the cups spin)
barometer	air pressure	falling pressure often means a storm is coming

> ⚠ **Stay safe**
>
> Put a thermometer in the **shade**, about 1.5 m above the ground. In direct sunshine it shows the temperature of the sunshine, not the air!

Reading the clouds



cumulus
puffy heaps: fine weather

stratus
flat grey sheet: dull, drizzle

cirrus
high thin wisps: weather may change

cumulonimbus
tall storm tower: heavy rain, thunder

Four common cloud types.

Clouds give clues. **White, puffy** clouds mean fine weather. **Dark grey** clouds hold more water and often bring rain. Clouds that grow **tall** in the afternoon can become thunderstorms. If clouds move quickly, the wind is strong. A change of wind direction, falling pressure and very damp air can all mean the weather is about to change.

Make your own forecast

A **forecast** is a prediction of the weather. Scientists record measurements every day, look for **trends** (patterns over time), and compare them with what happened before. You can do the same with a class weather station.

Bangladesh connection

The **Bangladesh Meteorological Department** (BMD) measures the weather at stations all over the country and gives forecasts and storm warnings on radio, TV, mobile phones and the internet.

Key words

weather	what the air is like today: temperature, rain, wind, clouds
climate	the usual weather of a place over many years
forecast	a prediction of the weather
rain gauge	collects and measures rainfall
wind vane	shows wind direction
anemometer	measures wind speed
barometer	measures air pressure
humid	damp air with a lot of water vapour

FACT CHECK · B1 · WEATHER AND CLIMATE

Weather words and tools

★☆☆ Grade 4

Name: Class: Date:

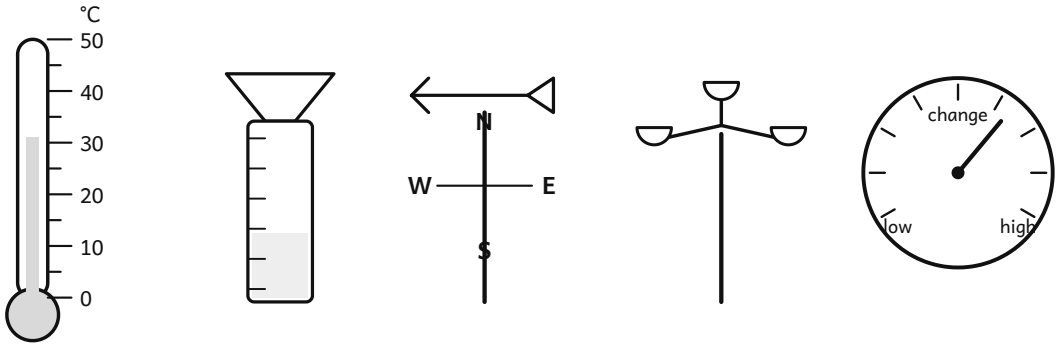
Learning goals

- Tell weather from climate.
- Match instruments to measurements.
- Name cloud types.

1 Weather or climate? Tick one. [4]

Sentence	Weather	Climate
It is very windy this afternoon.		
Sylhet gets a lot of rain every year.		
Tomorrow will be foggy.		
Bangladesh has a monsoon season every summer.		

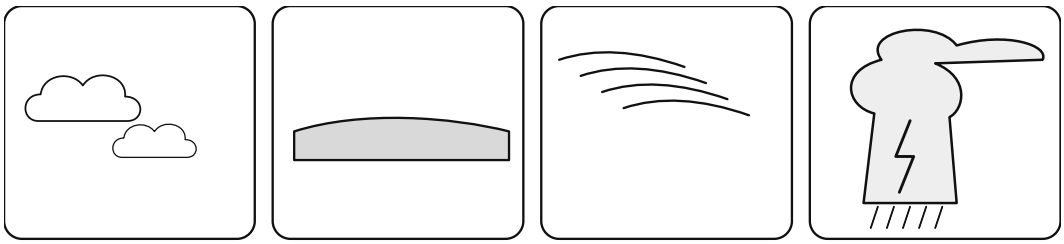
2 Write the name of each instrument and what it measures. [10]



A B C D E

Letter	Instrument	Measures
A		
B		
C		
D		
E		

3 **Name each cloud type.** [4]



A B C D

A	B	C	D

4 **Fill in the gaps.** [3]

WORD BANK

sun wind rises shade sinks

Warm air _____. Cool air moves in to take its place. Moving air is called _____. A thermometer should be kept in the _____.

Thinking about air and weather

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Predict warm and cool places.
- Explain air movement.
- Use clues to forecast.

1 **Rina measured the temperature in four places at school at 12 noon.** [4]

Place	Temperature (°C)
under a big tree	29
on the tin roof of the shed	45
in the classroom (fan on)	31
on the concrete playground in sun	38

a) Which was the warmest place? Explain why.

.....

.....

b) Which was the coolest place? Explain why.

.....

.....

2 **Why do many old houses in Bangladesh have small openings high up in the walls?** [2]

.....

.....

3 **The barometer is falling, the wind has changed direction and tall dark clouds are growing. What weather would you forecast? Why?** [3]

.....

.....

.....

4 **Draw a sea breeze. Label the warm land, the cool sea and arrows to show how the air moves.** [3]

Sea breeze

INVESTIGATE · B1 · WEATHER AND CLIMATE

A week at our weather station

★★☆
Grade 4

Name: Class: Date:

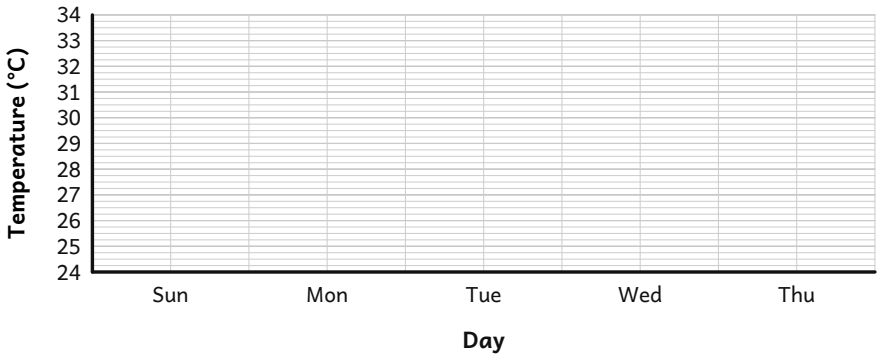
Learning goals

- Record weather data.
- Draw a line graph.
- Make a forecast from trends.

Class 4A recorded the weather at 9 o'clock every morning for one week in June.

Day	Sun	Mon	Tue	Wed	Thu
Temperature (°C)	31	32	30	28	27
Rainfall (mm)	0	2	10	25	30
Wind direction	S	S	SE	SE	SE
Clouds	a few cumulus	more cumulus	grey sheet	dark, tall	dark, tall
Pressure	steady	falling	falling	falling	low

1 **Draw a line graph of the temperature.** [3]



2 **Describe the trend in rainfall during the week.** [2]

.....

.....

3 **Forecast the weather for Friday. Use at least two pieces of evidence.** [3]

.....

.....

.....

4 **Why did they measure at the same time every day?** [1]

.....

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LEARN · PLAY · WONDER

Harvest before the flood: forecasts for haor farmers

★★★ Grade 4

Name: Class: Date:

Learning goals

- See how forecasts help people.
- Use rainfall data.
- Weigh up choices.

Science · Farming · Maths · Technology

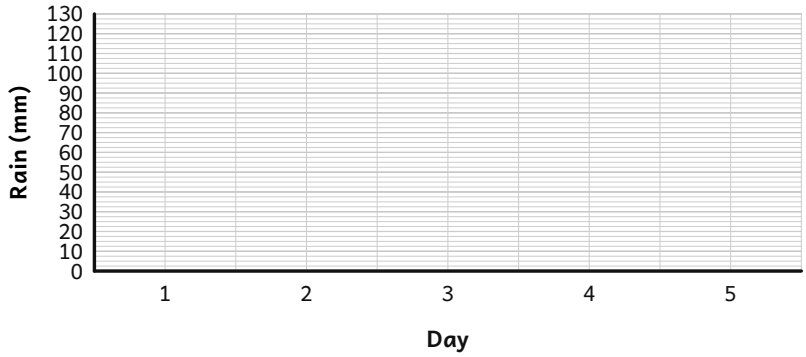
In the **haor** wetlands of the north-east, farmers grow **boro rice** in the dry season. The rice is ready to harvest in **April and May**. But heavy rain in the nearby hills can send a sudden **flash flood** rushing into the haor in just a day or two, drowning the whole crop.

Today, the **Bangladesh Meteorological Department** and the flood forecasting centre send warnings by radio, TV, mobile phone messages and local announcements. If a farmer knows heavy rain is coming, they can harvest early, even if the rice is not completely ripe, and save most of the crop.

Based on information about flash floods in the haors and early-warning services in Bangladesh. Data below are example values.

1 **Forecast rainfall in the hills for the next five days (example).** [3]

Day	1	2	3	4	5
Rain forecast (mm)	5	10	60	120	80



a) Draw the bar chart.

b) On which day should the farmer start harvesting? Explain.

2 **A farmer says, "My rice needs one more week to ripen." What should they do? Give reasons for your answer.** [2]

3 **Why are mobile phone messages a good way to warn farmers?** [2]

4 **Design a simple sign that a village could put up to show the flood risk this week (low, medium, high).** [2]

Flood risk sign

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	B	W	K	P	J	K	R	A	I	N
C	L	I	M	A	T	E	N	W	P	G
H	G	F	W	D	M	O	N	E	J	Y
C	I	A	F	O	R	E	C	A	S	T
M	K	M	R	M	C	F	B	T	J	U
D	G	P	O	U	O	H	P	H	E	V
V	A	J	U	N	D	J	T	E	J	U
A	U	D	J	T	S	B	U	R	R	N
N	G	O	J	M	W	O	W	I	N	D
E	E	C	L	O	U	D	O	H	T	A
W	D	E	H	N	A	Y	B	N	G	T

climate cloud forecast gauge monsoon rain vane weather wind

2 Which instrument measures wind speed? [1]

- ◯ barometer
- ◯ anemometer
- ◯ thermometer
- ◯ rain gauge

3 Which cloud is most likely to bring a thunderstorm? [1]

- ◯ cirrus
- ◯ cumulus
- ◯ cumulonimbus
- ◯ stratus

4 Design a simple rain gauge from a plastic bottle. Draw and label it. [3]

My rain gauge

5 **Challenge. Explain how wind is made. Use the words warm, rises and cool.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain weather and climate	○	○	○
predict warm and cool places	○	○	○
explain how warm and cool air move	○	○	○
use weather instruments	○	○	○
name four cloud types	○	○	○
make a forecast from a week of data	○	○	○